Robust Estimation for Markowitz

Quantitative Trading with R

Prof: Patrick Kerl

Group: Ridhima Lawane - 1779783

Abhiruchi Bharambe - 1783040

Shubham Darekar – 1783774

Table of Contents

Abstract.....3

1. Introduction.....3

2. Data and Model.....4

    2.1 Data.....4

    2.2 Unconstrained Mean-Variance Optimization.....4

    2.3 Constrained Mean-Variance Optimization.....4

    2.4 Robust Mean-Variance Optimization.....4

3. Implementation.....5

4. Results and Findings.....5

    4.1 Portfolio Comparison.....5

    4.2 Portfolio Weights.....6

    4.3 Sensitivity Analysis.....7

    4.4 Efficient Frontier.....8

5. Conclusion.....9

References.....9

Abstract

This seminar paper compares three portfolio optimization approaches: unconstrained Mean-Variance Optimization (MVO), constrained MVO, and robust MVO. The analysis is based on daily historical data of ten U.S. stocks from 2019 to 2023. In the unconstrained model, short selling is allowed and there is no maximum limit on individual stock weights. The constrained model does not allow short selling and limits each stock to a maximum weight of 20%. The robust model uses the same practical constraints while also considering uncertainty in return and risk.

The three approaches are compared using portfolio weights, annualized return, annualized volatility, and the Sharpe ratio. A sensitivity analysis is also performed by changing the maximum volatility allowed for the robust portfolio. The results show that introducing constraints causes clear changes in the portfolio weights. For the data used in this project, the robust portfolio gives an annualized return of 21.51%, volatility of 19.62%, and a Sharpe ratio of 1.096. The sensitivity analysis also shows that allowing higher portfolio volatility generally leads to a higher achievable return.

1. Introduction

Mean-Variance Optimization (MVO) is a commonly used method for portfolio selection. The main idea is to construct a portfolio by considering both expected return and risk. However, a basic MVO model can sometimes produce portfolios that may not be practical. For example, it can assign a very large weight to a single stock or use negative weights, which represent short selling.

To make the portfolio more realistic, additional constraints can be introduced. In this project, a constrained MVO model is used where short selling is not allowed and the maximum investment in one stock is limited to 20%. These constraints help to avoid very large individual positions and create a more diversified portfolio.

Another issue with portfolio optimization is that expected returns and risks are estimated using historical data. Therefore, these estimates are not known with certainty. To consider this problem, a robust MVO model is also implemented. The robust model includes uncertainty in the optimization while keeping the same long-only and 20% maximum-weight constraints.

The aim of this project is to compare the unconstrained, constrained, and robust MVO approaches and study how the different assumptions affect portfolio allocation and performance. The models are compared using portfolio return, volatility, Sharpe ratio, and individual stock weights. In addition, a sensitivity analysis is carried out by changing the maximum allowed robust volatility while keeping the other constraints fixed.

2. Data and Model

2.1 Data

The analysis uses adjusted daily closing prices for ten U.S. stocks: AAPL, MSFT, JPM, BAC, JNJ, PFE, XOM, CVX, AMZN, and WMT. The data cover the period from 1 January 2019 to 1 January 2024 and are downloaded from Yahoo Finance in R. Daily returns are calculated from the adjusted prices. The sample mean return vector and covariance matrix are then estimated from these daily returns. Annual values are obtained using 252 trading days.

2.2 Unconstrained Mean-Variance Optimization

The first model is a minimum-variance portfolio. It minimizes portfolio variance while requiring the portfolio weights to sum to one. No restriction is placed on individual weights, so negative weights and large positions are possible.

$$\text{minimize} \quad w'\Sigma w$$

$$\text{subject to} \quad \Sigma w_i = 1$$

2.3 Constrained Mean-Variance Optimization

The second model adds practical restrictions to the minimum-variance problem. Short selling is not allowed and every stock is limited to a maximum weight of 20%. The constraints are:

$$\Sigma w_i = 1, \quad 0 \le w_i \le 0.20$$

These restrictions prevent negative positions and reduce excessive concentration in a single stock.

2.4 Robust Mean-Variance Optimization

The robust model keeps the long-only and 20% maximum-weight restrictions but also includes uncertainty. The objective maximizes a worst-case return measure. In the implementation, the return uncertainty parameter ρ is based on the 20th percentile of the absolute estimated mean returns and is applied to all assets. The risk measure combines the usual covariance-based portfolio risk with an additional uncertainty penalty controlled by γ = 0.02.

$$\text{maximize } \mu'w - \rho'|w|$$

A maximum robust-volatility limit also restricts the robust portfolio. A 20% limit is used when feasible. This allows the model to search for a portfolio with a good return while taking estimation uncertainty and practical allocation limits into account.

3. Implementation

The models are implemented in R. The quantmod package is used to download adjusted stock prices and PerformanceAnalytics is used to calculate returns. The optimization problems are formulated and solved with CVXR. ggplot2 and reshape2 are used for the graphical comparison of portfolio weights, sensitivity results, and efficient frontiers.

For each portfolio, annualized return and volatility are calculated from the estimated daily returns and covariance matrix. The Sharpe ratio used in the comparison is calculated as annualized return divided by annualized volatility. The implementation also checks the minimum feasible robust risk before solving the robust portfolio.

For the sensitivity analysis, the maximum allowed robust volatility is changed while the long-only and 20% maximum-weight constraints remain fixed. This makes it possible to observe how the achievable return changes when the portfolio is allowed to take more risk. The constrained and robust efficient frontiers are also calculated over the same risk-limit range to make the graphical comparison consistent.

4. Results and Findings

4.1 Portfolio Comparison

Table 1 summarizes the main results. The unconstrained portfolio has the lowest volatility, but it contains short positions and has a maximum absolute stock weight of 42.34%. After adding practical constraints, the constrained portfolio has a return of 12.99% and volatility of 17.78%. For this sample, the robust portfolio gives the highest nominal return and Sharpe ratio, with a return of 21.51%, volatility of 19.62%, and Sharpe ratio of 1.096.

Portfolio	Return (%)	Risk (%)	Sharpe	Max Weight (%)	Shorts
Unconstrained	10.86	16.45	0.660	42.34	Yes
Constrained	12.99	17.78	0.731	20.00	No
Robust	21.51	19.62	1.096	20.00	No

Table 1: Comparison of the three portfolio optimization approaches.

4.2 Portfolio Weights

The portfolio weights change considerably when the practical constraints are introduced. For example, JNJ has a weight of 42.34% in the unconstrained portfolio but is reduced to 20% in the constrained portfolio because of the maximum-weight restriction. WMT is similarly reduced from 32.01% to 20%. Negative positions such as BAC (-15.01%) and CVX (-4.97%) become zero because short selling is not allowed.

The robust solution changes the allocation further. AAPL receives 20%, MSFT about 19.14%, while PFE falls to about 3.83% and AMZN receives zero weight. This shows that robustness can change portfolio composition even when the same long-only and maximum-weight restrictions are applied.

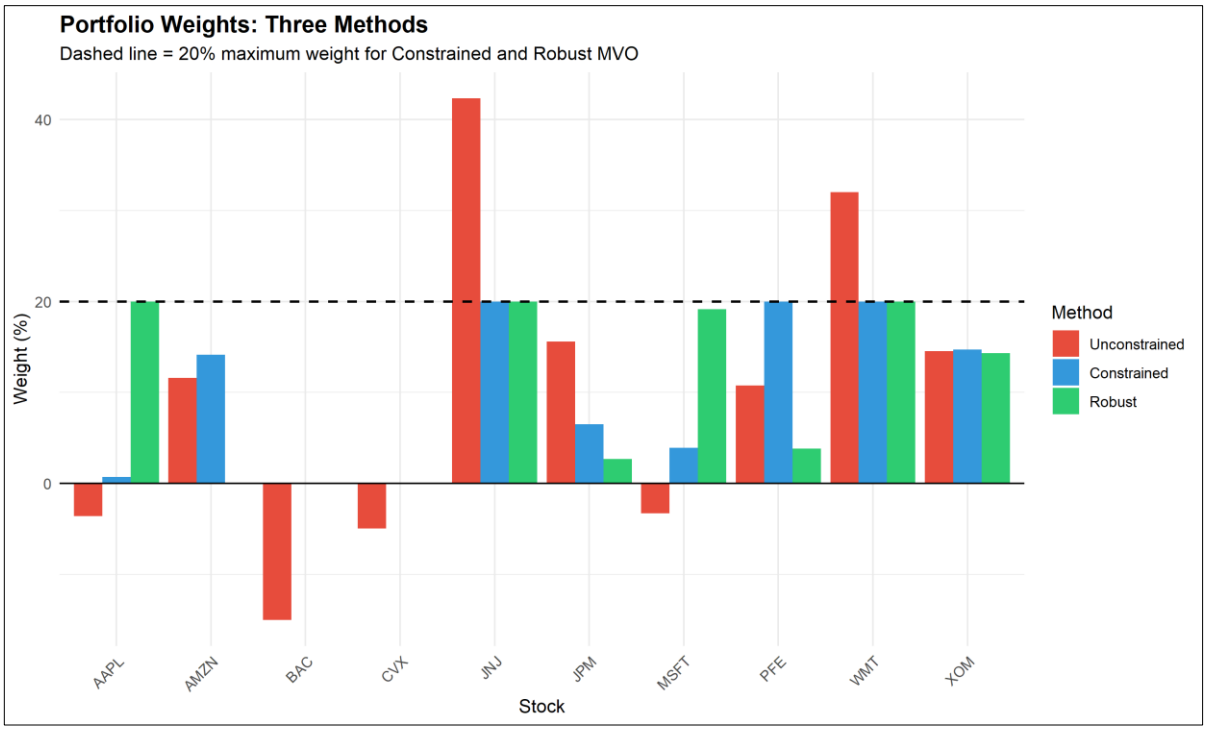


Figure 1: Portfolio weights under the three optimization approaches

4.3 Sensitivity Analysis

The sensitivity analysis varies the maximum robust-volatility constraint from 18.29% to 25.08%. The resulting portfolio returns increases from 14.57% to 25.36%. This confirms the expected risk-return relationship: when the model is allowed to accept more volatility, it can select portfolios with higher expected return.

Volatility Limit (%)	Actual Risk (%)	Robust Risk (%)	Return (%)	Sharpe
18.29	17.89	18.29	14.57	0.815
19.65	19.26	19.65	20.49	1.064
21.01	20.61	21.01	23.12	1.122
22.37	21.95	22.37	24.36	1.110
23.72	23.30	23.72	24.92	1.070
25.08	24.65	25.08	25.36	1.029

Table 2: Sensitivity of the robust portfolio to the volatility constraint.

The increase in return becomes smaller at the higher volatility limits. The Sharpe ratio reaches its highest value of 1.122 at a volatility limit of approximately 21.01% and then declines. Therefore, allowing more risk continues to increase nominal return, but it does not continuously improve risk-adjusted performance.

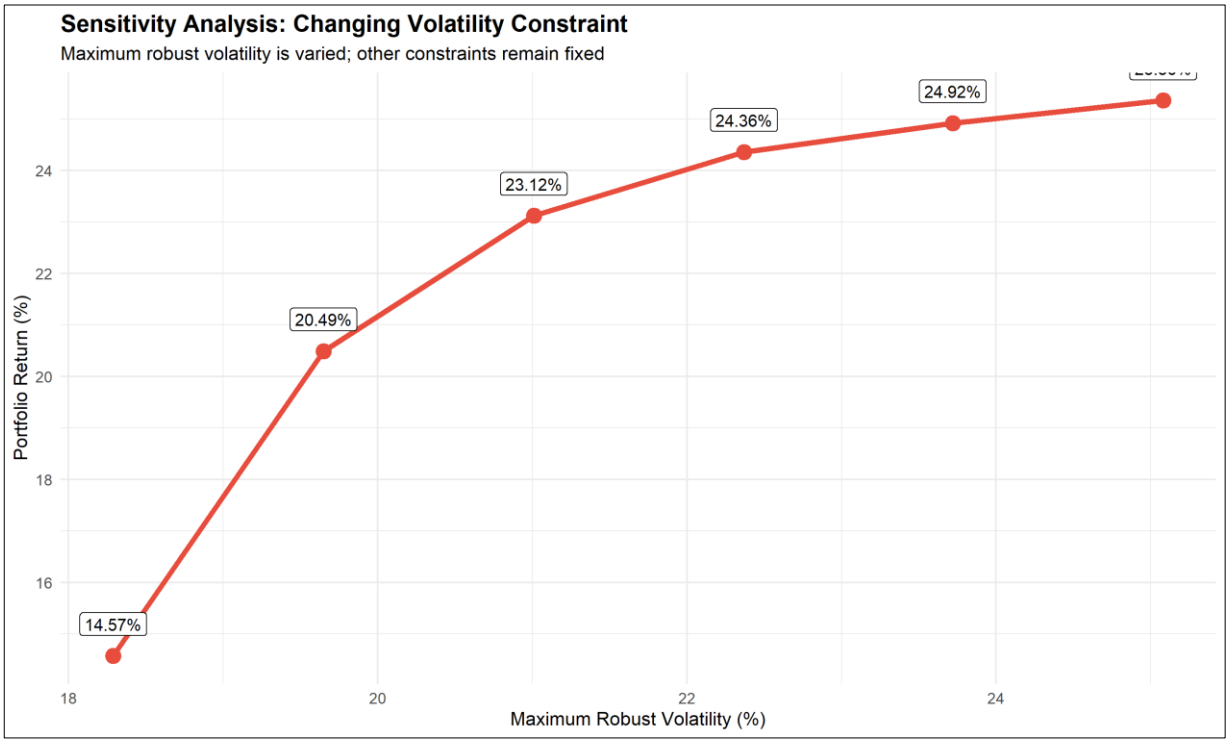


Figure 2: Sensitivity of portfolio return to the robust volatility constraint,

4.4 Efficient Frontier

The constrained and robust efficient frontiers are close to each other over much of the tested range. This does not mean that the robust model has no effect. The weight comparison shows that the allocation can change considerably even when the nominal risk-return curves remain similar. For this dataset and uncertainty specification, robustness therefore has a stronger visible effect on portfolio composition than on the overall shape of the nominal efficient frontier.

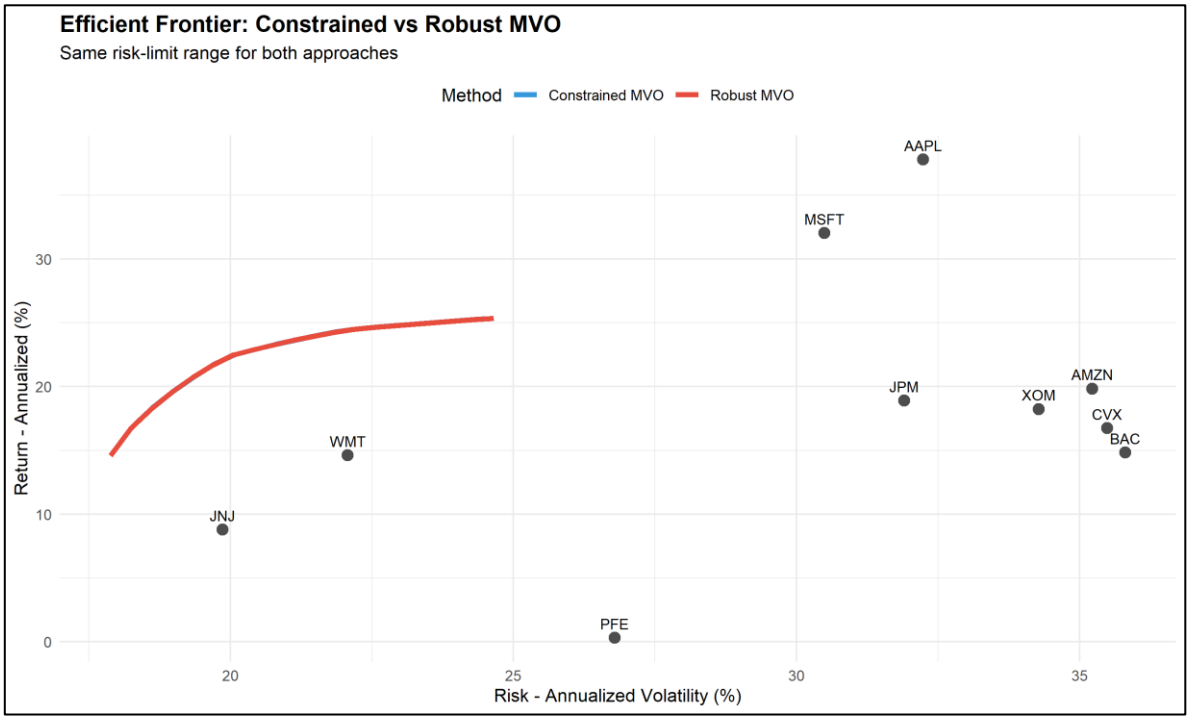


Figure 3: Constrained and robust efficient frontiers

5. Conclusion

This project compared unconstrained, constrained, and robust Mean-Variance Optimization using ten U.S. stocks. The results show that adding practical constraints has an important effect on portfolio allocation. The 20% maximum weight limit reduces highly concentrated positions, while the long-only condition removes negative portfolio weights.

For the historical sample used in the analysis, the robust portfolio achieved a nominal annualized return of 21.51%, volatility of 19.62%, and a Sharpe ratio of 1.096. These results should be interpreted as sample-specific and do not imply that robust MVO will always outperform the other approaches.

The sensitivity analysis shows that relaxing the volatility constraint increases the achievable return, but the additional benefit becomes smaller at higher risk levels. The Sharpe ratio also reaches a maximum before the highest tested volatility limit. Overall, the project demonstrates that portfolio constraints and uncertainty assumptions can have a clear effect on both portfolio weights and the risk-return trade-off.

References

- Markowitz, H. (1952). Portfolio Selection. The Journal of Finance, 7(1), 77-91.
- Boyd, S., & Vandenberghe, L. (2004). Convex Optimization. Cambridge University Press.
- CVXR. R package for disciplined convex optimization. Used for the optimization implementation.
- Yahoo Finance. Historically adjusted price data for the selected stocks, accessed through the quantmod R package.

Appendix

The complete R script used for data collection, portfolio optimization, sensitivity analysis, and figure generation is submitted separately with the seminar paper.